

AP data imputation1

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```
[1]: import numpy as np
import pandas as pd
import os
```

Import Amman Data

```
[2]: folder = r"C:\Users\user\OneDrive\Desktop\new ap mort\AP data\Amman/" #Define
      ↳the directory of folder containing Amman Stations
files = [file for file in os.listdir(folder) if file.endswith('.xlsx')]
amm = [pd.read_excel(folder + file) for file in files]
files = [file[2:5] for file in files]
for i in range(len(files)):
    amm[i]['Sta'] = files[i] #add a column containing each station's symbol
```

Import Irbid Data

```
[3]: folder = r"C:\Users\user\OneDrive\Desktop\new ap mort\AP data\Irbid/" #Define
      ↳the directory of folder containing Irbid Stations
files = [file for file in os.listdir(folder) if file.endswith('.xlsx')]
irb = [pd.read_excel(folder + file) for file in files]
files = [file[2:5] for file in files]
for i in range(len(files)):
    irb[i]['Sta'] = files[i] #add a column containing each station's symbol
```

Import Zarqa Data

```
[4]: folder = r"C:\Users\user\OneDrive\Desktop\new ap mort\AP data\Zarqaa/" #Define
      ↳the directory of folder containing Zarqa Stations
files = [file for file in os.listdir(folder) if file.endswith('.xlsx')]
zar = [pd.read_excel(folder + file) for file in files]
files = [file[2:5] for file in files]
for i in range(len(files)):
    zar[i]['Sta'] = files[i] #add a column containing each station's symbol
```

Concatenate the data of each city

```
[5]: amm = pd.concat(amm, ignore_index=True)
irb = pd.concat(irb, ignore_index=True)
zar = pd.concat(zar, ignore_index=True)
```

Define the `impute` function

```
[ ]: def impute(df, col):  
    """  
    Impute missing values in the specified column of the DataFrame using a  
    ↪ratio-based method.  
  
    df: pandas DataFrame containing the data  
    col: string, the name of the column to impute  
  
    """  
    for day in df['Date'].unique():  
        year = pd.to_datetime(day).year  
        sdf = df[df['Date'] == day]  
        nandf = sdf[sdf[col].isna()]  
        if not nandf.empty:  
            mean_value = sdf[col].mean()  
            if len(nandf) == 1:  
                sta_ann_mean = df[(df['Sta'] == nandf['Sta'].values[0]) &  
↪(df['Date'].dt.year == year)][col].mean()  
                rest_ann_mean = df[(df['Sta'] != nandf['Sta'].values[0]) &  
↪(df['Date'].dt.year == year)][col].mean()  
                df.loc[nandf.index, col] = mean_value * (sta_ann_mean /  
↪rest_ann_mean)  
            if len(nandf) > 1:  
                for i in range(len(nandf)):  
                    sta_ann_mean = df[(df['Sta'] == nandf['Sta'].values[i]) &  
↪(df['Date'].dt.year == year)][col].mean()  
                    rest_ann_mean = df[(df['Sta'] != nandf['Sta'].values[i]) &  
↪(df['Date'].dt.year == year)][col].mean()  
                    df.loc[nandf.index[i], col] = mean_value * (sta_ann_mean /  
↪rest_ann_mean)  
  
    return None
```

For PM10, check # of missing values for each city

```
[17]: print(f"Amman: {len(amm[amm['PM10'].isna()])}")  
       print(f"Irbid: {len(irb[irb['PM10'].isna()])}")  
       print(f"Zarqa: {len(zar[zar['PM10'].isna()])}")
```

Amman: 480

Irbid: 680

Zarqa: 470

Perform the imputation for PM10

```
[20]: impute(amm, 'PM10')
      impute(irb, 'PM10')
      impute(zar, 'PM10')
```

Check # of PM10 missing values after imputation

```
[21]: print(f"Amman: {len(amm[amm['PM10'].isna()])}")
      print(f"Irbid: {len(irb[irb['PM10'].isna()])}")
      print(f"Zarqa: {len(zar[zar['PM10'].isna()])}")
```

Amman: 180

Irbid: 400

Zarqa: 126

Similarly for NO2

```
[22]: print(len(amm[amm['NO2'].isna()]))
      print(len(irb[irb['NO2'].isna()]))
      print(len(zar[zar['NO2'].isna()]))
```

573

188

140

```
[23]: impute(amm, 'NO2')
      impute(irb, 'NO2')
      impute(zar, 'NO2')
```

```
[24]: print(len(amm[amm['NO2'].isna()]))
      print(len(irb[irb['NO2'].isna()]))
      print(len(zar[zar['NO2'].isna()]))
```

192

66

62